UNIT 5 · 9 hours

Operational Amplifier: Introduction and the Ideal Op-Amp



Op-Amp and Linear IC Applications

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

- Describe the op-amp symbol and terminals
- State the properties of an ideal op-amp
- Explain why negative feedback is used
- Compare ideal and practical (741) op-amps

Syllabus context

Introduction, op-amp specifications, differential and common-mode operation, op-amp applications: inverting/non-inverting amplifier, summing, integrator, differentiator, comparator, 555 timer IC in astable mode, 78XX series IC voltage regulators.

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What is an operational amplifier?

High-gain DC-coupled amplifier with differential inputs and a single output.

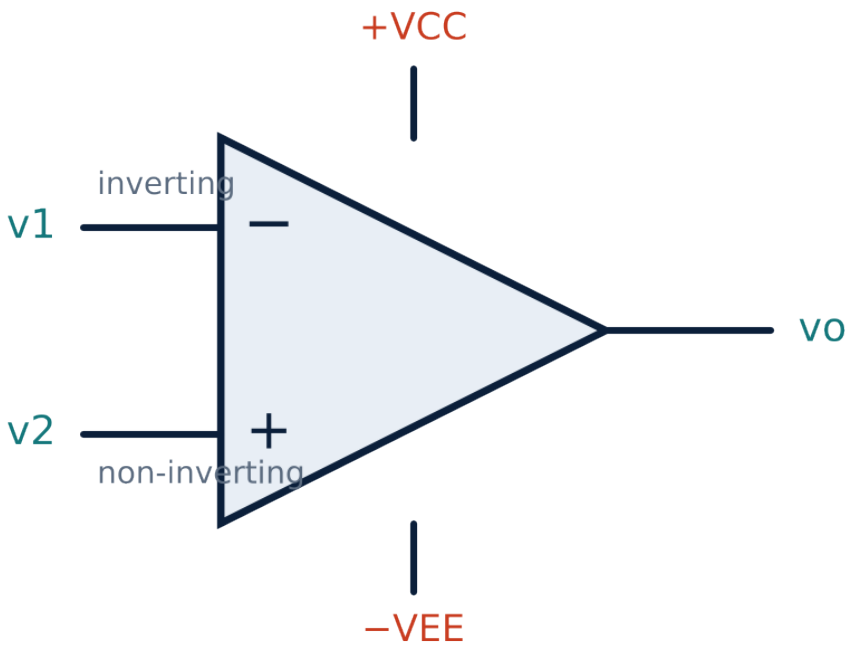
Originally built to perform mathematical operations (sum, integrate, differentiate) in analog computers.

IC form (1965): low cost, small, reliable; the popular type is the 741.

Uses: amplifiers, filters, comparators, converters, oscillators, instrumentation.

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Symbol and terminals



- Inverting input (−), non-inverting input (+), output.
- Power supply +VCC and −VEE (e.g. ±15 V).
- Output: vo = A (v2 − v1).
- A = open-loop gain (no feedback).

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Ideal op-amp properties

Infinite gain

$A \to \infty$; hence $v_1 \approx v_2$ (virtual short).

Infinite input resistance

No current enters the inputs: $i_1 = i_2 = 0$.

Zero output resistance

Output is an ideal voltage source.

Also: infinite bandwidth, infinite CMRR, zero offset voltage, infinite slew rate.

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Ideal versus practical op-amp (741)

Parameter	Ideal	Practical (741)
Open-loop gain	Infinite	≈ 2×10^5
Input impedance	Infinite	≈ 2 MΩ
Output impedance	Zero	≈ 75 Ω
Bandwidth (unity gain)	Infinite	≈ 1 MHz
Slew rate	Infinite	0.5 V/µs
CMRR	Infinite	70–120 dB
Output swing	±VCC	≈ ±13 V for ±15 V

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Negative feedback

- Feeds part of the output back to the inverting input.
- Closed-loop gain depends only on external resistors, not on the op-amp's A.
- Improves stability, bandwidth, linearity and input/output impedance.
- Reduces the gain from A (~10^5) to a controlled value.

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Key takeaways

- Op-amp: high-gain differential amplifier, vo = A(v2 − v1)
- Ideal: A = ∞, Rin = ∞, Ro = 0, no input current
- Negative feedback makes gain depend on external resistors
- 741: A ≈ 2×10^5, 1 MHz, 0.5 V/µs

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Practice questions

Q1. List the properties of an ideal op-amp.

Q2. Why is negative feedback used?

Q3. What are the five terminals of an op-amp IC?

Q4. Compare ideal and practical op-amps.